

Quantifying effects of soil heterogeneity on groundwater pollution at four sites in USA

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Abstract Four sites located in the north-eastern region of the United States of America have been chosen to investigate the impacts of soil heterogeneity in the transport of solutes (bromide and chloride) through the vadose zone (the zone in the soil that lies below the root zone and above the permanent saturated groundwater). A recently proposed mathematical model based on the cumulative beta distribution has been deployed to compare and contrast the regions' heterogeneity from multiple sample percolation experiments. Significant differences in patterns of solute leaching were observed even over a small spatial scale, indicating that traditional sampling methods for solute transport, for example the gravity pan or suction lysimeters, or more recent inventions such as the multiple sample percolation systems may not be effective in estimating solute fluxes in soils when a significant degree of soil heterogeneity is present. Consequently, ignoring soil heterogeneity in solute transport studies will likely result in under- or over-prediction of leached fluxes and potentially lead to serious pollution of soils and/or groundwater. The cumulative beta distribution technique is found to be a versatile and simple technique of gaining valuable information regarding soil heterogeneity effects on solute transport. It is also an excellent tool for guiding future decisions of experimental designs particularly in regard to the number of samples within one site and the number of sampling locations between sites required to obtain a representative estimate of field solute or drainage flux.

Keywords: solutes, drainage, cumulative beta distribution, pollution, heterogeneity index, bromide, chloride, leaching, multiple sample percolation system, preferential flow.

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The movement of chemicals through agricultural soils has long been a topic of considerable theoretical and experimental research due to concerns for the quality of groundwater and surface waters^[1]. In many cases, serious human and stock health implications are associated with this form of pollution^[2]. The chemi-

cals of interest include nutrients, pesticides, salts and industrial wastes^[3–5]. In the case of transport of pathogens^[6] and heavy metals^[7–12], serious human and environmental health risk may occur. In the case of nutrients^[13–15], leaching losses also represent a decline in soil fertility with economic consequences, and in

the case of nitrate leaching, reduction of productivity due to soil acidification. In the case of pesticides and industrial wastes^[16–19], small but highly toxic chemicals can be transported to groundwater and remain there for hundreds and even thousands of years. Studying the transport of the fertilisers and pesticides in the vadose zone (i.e. the zone below the root zone and above the saturated groundwater) provides a first prediction of solute leaching and hence potential risk of groundwater pollution.

Monitoring and modelling solute transport is complicated by non-random spatial and temporal variations of physical, chemical and biological components of soils^[20]. One manifestation of spatial and temporal heterogeneity in agricultural and natural soils is the phenomenon of preferential flow, a general term used to describe a variety of physical and chemical non-equilibrium flow processes^[1,20]. In fine-textured soils, and soils containing a large percentage of clay and organic matter, large and more or less continuous macropores resulting from soil shrinkage and swelling, earthworm and animal burrows, decayed root holes and subsurface erosion, operate as preferential flow paths and cause rapid movement of solutes through the unsaturated zone^[21,22]. Preferential flow is not confined solely to heavy clay soils, although these represent the worst case scenario. Preferential flow phenomena have also been widely observed in many sandy soils in the field^[23–26]. So, regardless of the soil, clay or sand, preferential flow results in rapid transport of solutes through the unsaturated zone.

Many solute transport studies in the past have used homogenised field soils in laboratory studies. However, laboratory experiments based on repacked, sterile, homogenised soil cores bear little resemblance to physical reality as they destroy the soil structure and under-estimate the solute loss and risk of contamination to groundwater reserves^[20]. A better knowledge of the factors causing non-equilibrium and heterogeneous solute transport is required for sustainable ecological management of chemicals in agricultural soils.

Much past research has shown that soil hetero-

geneity, soil structure and water repellency of soils are major factors resulting in preferential flow^[23–26]. A more accurate prediction of soil heterogeneity and the occurrence of preferential flow paths are important to effectively model solute leaching. In the past two decades, several approaches have been developed to monitor the temporal and spatial patterns of solutes and water in the soil. A currently widely used method to monitor solute leaching in the vadose zone under field conditions is the *Multiple Sample Percolation System* (MSPS)^[1]. The Multiple Sample Percolation System (MSPS) has been widely used in Australia, Europe and the USA for ascertaining pesticide, nutrients, and other soluble contaminant transport in soils^[4,20]. An MSPS typically consists of 25 to 300 individual collecting cells (or mini catchments) placed under an undisturbed soil sample core. Each mini-catchment collects water and solutes leaching from the undisturbed soil cores. Experimental details of MSPS have been previously described^[3,4].

Modelling the soil heterogeneity can be conducted in both space and time^[28,29]. The temporal distribution of solute leaching can be described by the classical breakthrough curve^[28]. For the variability in space, however, a recently developed simple, robust, mathematical model, based on the beta-statistical distribution, can be used to quantify the degree of soil heterogeneity^[29,30]. The method relies on fitting two parameters to the plot a cumulative beta distribution curve using data generated from MSPS experiments. An advantage of this method to previous methods is that no knowledge of the soil structure is required to develop an indication of spatial heterogeneity, just observed findings generated from multiple sample leaching experiments. Hence this approach has been demonstrated to be a very powerful method in pollution studies^[30].

In this paper, we present the analysis of the application of this new model to a comprehensive data set collected previously in the United States. We demonstrate the utility and versatility of the model.

1 Materials and methods

1.1 Experimental design

Multiple sample percolation systems (MSPS) are an excellent means to obtain accurate values of the solute and water flux in soil experiments and have been shown to effectively represent the impacts of soil heterogeneity on these fluxes^[1,4,20].

The MSPS data of the American soils were collected for six experiments at four sites: (i) University of Delaware, New York (‘Delaware site’), (ii) Cornell University’s Thompson Vegetable Farm near Freeville, New York (‘Freeville site’), (iii) Cornell University’s Experimental Research Farm in Willsboro, New York (‘Willsboro site’), (iv) and Cornell University’s orchard in Ithaca, New York (‘Orchard site’). Two experiments were carried out at the Freeville site (‘Freeville-1’ and ‘Freeville-2’) and two experiments at the Delaware site (‘Delaware-1’ and ‘Delaware-2’) (fig. 1). The data were collected by Dr Jan Boll as part of his doctoral thesis at Cornell University, New York^[1]. Two different MSPS samplers were used: (i) gravity and (ii) wick pan samplers. These were installed at 0.6–0.7 m depth (fig. 2). Each sampler consisted of 25 cells, each 6 by 6 cm arranged in a square grid, which were sampled separately (for further details on samples see^[1,4,5]). In each experiment a pulse of a non-adsorbed tracer (bromide or chloride) was applied and was followed by irrigation or natural rainfall. A summary of the field experiments is given in table 1.

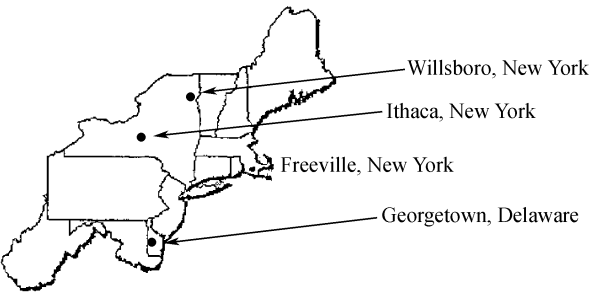


Fig. 1. Location of experimental sites, United States of America^[5].

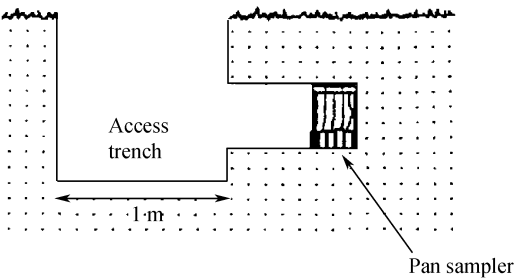


Fig. 2. Sampling setup^[5].

1.2 Mathematical modelling

The recently developed method^[4], based on the beta-statistical distribution, is used to quantify the degree of soil heterogeneity for the samplers collected at the four sites in the United States of America. The method relies on fitting the two shape parameters, α and ζ , of the cumulative beta distribution to leaching data generated from the MSPS. In this case, the cumulative beta distribution is fitted to data collected

Table 1 Summary of experimental design^[1,5]

	Freeville 1	Freeville 2	Orchard	Delaware-1	Delaware 2	Willsboro
Soil type	silt loam	silt loam	silty clay loam	sandy loam	sandy loam	sandy clay loam
Type of tracer	bromide	bromide	bromide	chloride	bromide	chloride nitrate
Type of pulse application	solution	solution	solution	solution	solution	flakes
Pulse concentration	7.8×10^{-3} kg/L	14.5×10^{-3} kg/L	7.8×10^{-3} kg/L	2.2×10^{-3} kg/L	1×10^{-2} kg/L	4000 kg/ha Cl 23 kg/ha N
Pulse length/cm	4	4	3.5	4	2	NA
Type of water application	sprinkler	sprinkler	sprinkler	sprinkler	natural rainfall	sprinkler
Duration of rainfall	2–3 h/d	13.5 h	2–3 h/d	6 h: 10 min on, 35 off	variable	2 events: 7, 4.5 h
Rainfall rate/cm·h ⁻¹	1.5	1.5	1.5	1.5	variable	0.9
Length of study/d	21	1	21, 12	21	131	2
Total applied water/cm	84	20	49, 35	63	40	15
Depth of sampler/cm	60	60	60	60–70	60–70	60
Sampler type (number)	Wick (2), gravity (2)	Wick (2), gravity (2)	Wick (2), gravity (2)	Wick (2)	Wick (2)	Wick (4)

from the 25 mini-catchments in each of the 6 experiments across the 4 different sites over a period of several months. Full details are explained elsewhere^[2,5], however a simple example is provide here. Consider the drainage flux for example. Water leaching from each mini-catchment in the MSPS is ranked in descending order and the fraction of cumulative leaching per mini-catchment is plotted with the cumulative fraction of total cross area per mini-catchment as shown in fig. 3. The dark solid line represents the fitted cumulative beta distribution and the dots represent the experimental data.

Once this plot has been generated and the optimal free parameters of the beta distribution have been determined then the degree of heterogeneity in the soil leaching pattern can be determined by calculating the Heterogeneity Index (HI) or the (relative) Integrated Area (IA)^[2]. The HI is a scaled ratio of the standard deviation over the mean of the cumulative beta distribution^[2]. The IA method is simply the area under the curve in fig. 3. Uniform or homogeneous flow of water or solutes is indicated when $HI = 1$ (i.e. when $\alpha = \zeta = 1$) or $IA = 0.5$ and heterogeneity is indicated by $HI > 1$ or $IA > 0.5$. Large values of HI or IA indicate preferential flow^[3,4]. There may be qualitative differences between the use of the HI and IA, and therefore both indices are used in this study.

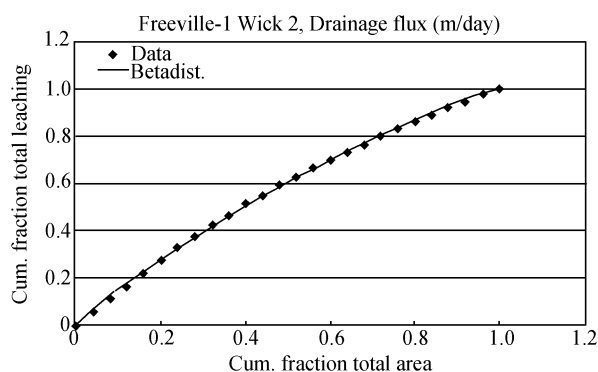


Fig. 3. Observed and predicted fractions of the drainage flux at the Freeville site (experiment 2) using a wick pan sampler plotted with the fraction of the cross-sectional area of each cell.

1.3 Field experimental site descriptions

Freeville site^[5]:

Soil: Well-drained Genesee silt loam (fine-sandy, mixed, non-acid, mesic Typic Udifluent) characterized by 0.6 m of dark brown loamy soil containing 30,000–50,000 worm and root channels per square meter in diameters ranging from 0.5 to 3 mm, overlying a dark brown silt loam to very sandy loam. At approximately 1.8 m layers of gravel and sand exist. The saturated hydraulic conductivity is between 35 and 120 cm/day.

Samplers: 2 wick and 2 gravity pan samplers on a grass plot.

Application Freeville-1 experiment: Twenty millimetres of artificial rainfall were applied twice a day (10 A.M. and 4 P.M.) starting on July 25, 1990. The rainfall rate was 10–15 mm/h. After a constant pan outflow was achieved, on August 2, a 7.8 g/L bromide solution was applied for 1 day, followed by 22 days of artificial rainfall as before.

Application Freeville-2 experiment: The plot was irrigated 3 times with 20 mm of water on June 24, 1991. The next day, pulse of 20 mm of water containing 14.5 g/L bromide was applied. The bromide pulse was followed immediately by a continuous water application of 11.5 h at a rate of 15–20 mm/h.

Delaware site^[5]:

Soil: The upper 0.6–0.8 m of the Evesboro sandy loam (mesic, coated, Typic Quartzipsamment) is a structureless, single, grain, yellowish-brown loamy sand with remains of roots from an old tree stand. Layers of fine sand (or occasionally greyish loam) and coarse material, typical of Pleistocene fluvial deposits, occur below 0.8 m depth. For the upper 0.3 m the average hydraulic conductivity is 4 m/day, and for the 0.3–0.6 m depth it is 2.5 m/day.

Samplers: 4 wick pan samplers: one under a ridge tillage plot (RT), two under reduced tillage and an application of poultry manure (PM1 and PM2), and one under reduced tillage with a winter cover of white clover (WC).

Application Delaware-1 experiment: The plots were irrigated twice with 40 mm of water and on July

16, 1992, a 40 mm of a solution with 2.2 g/L chloride was applied to each plot, followed by 21 days of artificial rainfall. The rain was applied from 6 A.M. until noon in cycles of 10 min on and 35 min off. The total amount of rain on the plots was 40 mm/day.

Application Delaware-2 experiment: This experiment was carried out from October 1992 to March 1993. A 10 g/L bromide solution in 20 mm of artificial rainfall was followed by approximately 400 mm of natural rainfall during 131 days.

Orchard site^[5]:

Soil: The Hudson silty clay loam consists of a pale brown fine-grained soil with a sub-granular blocky structure and hexagonal shaped peds of 0.2—0.3 m in diameter. Surface connected cracks were present, some as wide as 10—20 mm. For the upper 25 cm the hydraulic conductivity ranges from 40 to 120 cm/day and for the depths from 0.25 to 1.10 m the conductivity is between 1 and 40 cm/day. During summer there are higher measured conductivities, because of the flow of water through macropores.

Samplers: Two plots were used, a mowed grass covered plot and a moss covered plot. Each plot had one wick and one gravity pan sampler.

Application: On July 18, 1991, water was applied to the grass plot at a rate of 10—15 mm/h for 3—4 h. A pulse of 6 g/L bromide was added to the first irrigation. Irrigation was continued daily until August 12 for six days. The moss plot was irrigated daily (except three times) from July 29 until August 12. The first irrigation contained 6.8 g/L bromide.

Willsboro site^[5]:

Soil: The upper 0.4 m of the Rhinebeck Variant clay (illitic, mesic, Aeric Ochraqualfs) is greyish brown with a moderate medium granular structure penetrated by many roots. Lower of the profile, the structure became angular blocky with rocks present and fewer roots. In the 0—0.3 m soil layer, the average conductivity is 60 cm. At larger depths the conductivity increases again to approximately 60 cm/day. Water movement in the layer from 30—60 cm is

through macropores only.

Samplers: 2 wick pan samplers in a no-tillage plot (NT) and 2 wick pan samplers in a conventionally tilled plot (CT).

Application: 23 kg N/ha on May 5, 1993 and 4000 kg Cl/ha on August 17, 1993, were surface-applied followed by two 9 mm/h irrigation events on August 18 and 19, 1993, lasting 7 h on the first day and 4.75 h on the second day.

1.4 Descriptions of collected data sets

Freeville site^[5]:

Average drainage flux (mL/day), solute flux (cm/day) and dispersion coefficient (cm²/day) per cell for 2 wick pan samplers and 2 gravity pan samplers for Freeville 1 and Freeville 2 experiment.

Daily drainage flux (mg/L) and bromide concentration (mg/L) per cell for 2 wick pan samplers and 2 gravity pan samplers for Freeville-1 experiment.

Drainage flux (mg/L) per cell for 2 wick pan samplers and 2 gravity pan samplers 4 and 16 h after the 1st irrigation, 0 and 3 h after the 2nd irrigation, and 0 and 2 h after the 3rd irrigation for Freeville-1 experiment.

Delaware site^[5]:

Average drainage flux (mL/day), solute flux (cm/day) and dispersion coefficient (cm²/day) per cell at RT plot (RT = rich tillage), PM1 and PM2 plot (= reduced tillage), and at WC plot (reduced tillage, covered white clover) for Delaware 1 and Delaware 2 experiment.

Daily drainage flux (mg/L) and chloride concentration (mg/L) per cell for 2 wick pan samplers in RT, PM1, PM2 and WC plot for Delaware-1 experiment.

Orchard site^[5]:

Average drainage flux (mL/day), solute flux (cm/day) and dispersion coefficient (cm²/day) per cell for 2 wick pan samplers (1 moss covered plot, 1 grass covered plot) and 2 gravity pan samplers (1 moss covered plot, 1 grass covered plot).

Daily drainage flux (mg/L) and bromide concentration (mg/L) per cell for 2 wick pan samplers (1 moss covered plot, 1 grass covered plot) and 2 gravity pan samplers (1 moss covered plot, 1 grass covered plot).

Willsboro site^[5]:

Average drainage flux (mL/day), solute flux (cm/day) and dispersion coefficient (cm²/day) per cell for 2 wick pan samplers (1 tillage plot, 1 no tillage plot) and 2 gravity pan samplers (1 tillage plot, 1 no tillage plot).

Drainage flux (mL/h), chloride concentration (mg/L), nitrate concentration (mg/L) per cell per wick pan sampler (1 tillage, 1 no tillage) 0, 2, 15 and 18 h after the 1st irrigation, and 0, 2 and 18 h after the 2nd irrigation.

2 Results and discussion

The observed (monitored data) and predicted (using the cumulative beta distribution function) fractions of total leaching (drainage flux) collected at the four different experimental sites plotted with the actual fraction of total cross-sectional area, the HI, α , ζ and the IA are given in table 2. The fraction of the cross-sectional area through which 10%, 50% and 90% of the total drainage flux is leached, is also presented in table 2. Another important characteristic of solute transport is the point where the gradient of the curve is equal to 1. If the gradient is greater than 1, then the flow lines converge; if the gradient is less than 1 then this defines regions with diverging flow; if gradient is equal to 1, then flow lines run parallel (neither converging or diverging). The x -coordinate in table 2 is the fraction of the sample area with stronger preferential flow (converging flow paths); the y -coordinate is the fraction of total leaching (drainage) with stronger preferential flow (converging flow paths).

2.1 Freeville

At the Freeville site 2 wick and 2 gravity pan samplers were installed in a grass covered plot. Two experiments were carried out with different amounts and duration of rainfall (irrigation). Comparing the cumulative beta distribution curve and the curve char-

acteristics of both wick pan samplers presented in tables 2 and 3 shows that in the Freeville-1 experiment as well as in the Freeville-2 experiment wick pan sampler 2 has a smaller HI and IA than wick pan sampler 1.

To examine whether the IA of wick pan sampler 1 & 2 differ 'significantly', the daily drainage flux (over 22 days) of both wick pan samplers of experiment Freeville-1 was tested using a distribution-free, non-parametric statistical test (Wilcoxon Signed Ranks Test). The result shows a statistically significant difference between the distribution of wick pan sampler 1 and wick pan sampler 2 ($p < 0.05$). This result indicates that even within a small area (2 m²), significant heterogeneity exists in leaching patterns. This result has profound consequences as it indicates that neither the wick nor the gravity sampler cannot be used to represent solute movement over larger areas. To reliably estimate the risk of groundwater pollution, more wick pan samplers are required to effectively obtain a reliable average.

The average drainage flux collected in each cell over the 22 day period for the wick pan sampler 1 & 2 at the Freeville-1 site is presented in figs. 4 and 5. In these graphs the difference between in wick pan sampler 1 and wick pan sampler 2 is very clearly observed. Wick pan sampler 1, which has a higher HI and IA than wick pan sampler 1, has more cells with a higher drainage flux than wick pan sampler 2 eg cell 1 (with a mean of circa 350 mL/day) and cell 13 (with a mean of circa 250 mL/day) thus indicating more preferential flow.

Comparing the cumulative beta distribution characteristics of the gravity pan samplers at the Freeville site shows there is a small difference between gravity pan sampler 1 & 2. Comparing the cumulative beta distribution curve of the wick and gravity pan samplers shows that using gravity pan samplers in all cases results in a higher value of HI and IA. Boll^[1] (1995) concluded that in the field the wick pan samplers performed better than the gravity pan samplers, resulting in an overestimation of the degree of soil heterogeneity for the gravity samplers.

Table 2a Fitted beta distribution for gravity and wick pan sampler 1&2, Freeville-1&2

	Freeville-1 Gravity 1	Freeville-2 Gravity 1	Freeville-1 Gravity 2	Freeville-2 Gravity 2	Freeville-1 Wick 1	Freeville-2 Wick 1	Freeville-1 Wick 2	Freeville-2 Wick 2
HI	1.42	1.29	1.50	1.25	1.28	1.20	1.15	1.19
α	0.8183	0.7752	0.7696	0.8800	0.7353	0.8378	0.881	0.8144
ζ	2.2211	1.3300	2.4198	1.5772	1.1628	1.2253	1.1841	1.1377
IA	0.69	0.61	0.73	0.62	0.59	0.57	0.55	0.56
Gradient = 1 (x, y)	0.36, 0.70	0.52, 0.70	0.36, 0.75	0.52, 0.73	0.38, 0.55	0.40, 0.53	0.42, 0.53	0.40, 0.52
Frac. area through which x% leached								
10%	4%	4%	3%	5%	4%	6%	7%	6%
50%	20%	32%	18%	31%	33%	37%	39%	38%
90%	62%	78%	56%	76%	83%	81%	85%	84%

Table 2b Fitted beta distribution for gravity and wick pan sampler for sites PM1, PM2, RT and WC, Delaware-1&2

	Delaware-1 PM1	Delaware-2 PM1	Delaware-1 PM2	Delaware-2 PM2	Delaware-1 RT	Delaware-2 RT	Delaware-1 WC
HI	1.30	1.37	1.37	1.31	1.25	1.35	1.34
α	0.863	0.8707	0.7108	0.8346	0.857	0.9015	0.8538
ζ	1.7812	2.2115	1.3762	1.6982	1.4977	2.3256	1.941
IA	0.65	0.69	0.63	0.64	0.61	0.69	0.67
Gradient = 1 (x, y)	0.50, 0.75	0.40, 0.63	0.40, 0.72	0.48, 0.72	0.44, 0.63	0.48, 0.80	0.36, 0.65
Frac. area through which x% leached							
10%	4%	3%	4%	4%	4%	4%	3%
50%	28%	29%	22%	28%	32%	23%	26%
90%	70%	76%	64%	70%	76%	60%	67%

Table 2c Fitted beta distribution for gravity and wick pan sampler for grass/moss plot, Orchard

	Orchard Gravity grass	Orchard Wick grass	Orchard Gravity moss	Orchard Wick moss
HI	1.53	1.78	1.61	1.52
α	0.6308	0.5664	0.62642	0.7991
ζ	1.5763	2.3316	0.8936	2.8985
IA	0.68	0.77	0.84	0.75
Gradient = 1 (x, y)	0.32, 0.63	0.36, 0.81	0.26, 0.86	0.32, 0.75
Frac. area through which x% leached				
10%	3%	2%	2%	3%
50%	21%	12%	9%	15%
90%	70%	50%	31%	52%

Table 2d Fitted beta distribution for no-tillage and tillage plots, wick pan samplers, Willsboro

	Willsboro wick 1 no - till	Willsboro wick 2 no - till	Willsboro wick 3 conv till	Willsboro wick 4 conv till
HI	1.41	1.34	1.38	1.25
α	0.6688	0.8163	0.8711	0.9981
ζ	1.3338	1.7129	2.2871	2.1375
IA	0.64	0.65	0.69	0.66
Gradient = 1 (x,y)	0.40, 0.63	0.48, 0.74	0.44, 0.78	0.52, 0.79
Frac. area through which x% leached				
10%	2%	4%	3%	5%
50%	28%	26%	23%	27%
90%	74%	72%	62%	65%

Table 3 Average drainage flux HI and IA for wick pan samplers Freeville-1&2 experiment

	Wick 1, Freeville 1	Wick 2, Freeville 1	Wick 1, Freeville 2	Wick 2, Freeville 2
HI	1.28	1.15	1.20	1.19
IA	0.59	0.55	0.59	0.56

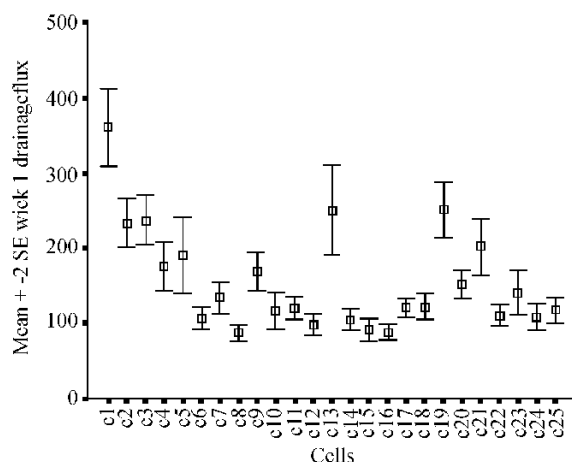


Fig. 4. Average drainage flux and standard errors for each collecting cell for wick pan sampler 1 over a 22 day period.

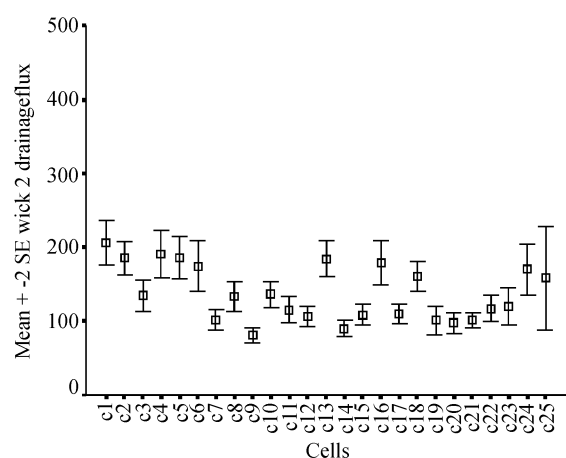


Fig. 5. Average drainage flux and standard errors for each collecting cell for wick pan sampler 2 over a 22 day period.

2.2 Delaware

At the Delaware site 2, wick pan samplers were installed: one in a plot under rich tillage (RT), 2 under reduced tillage (PM1 and PM2) and one under reduced tillage plot with a winter cover crop of rye and white clover (WC). Two experiments were carried out at

each plot; one with natural rainfall and in the other experiment was irrigated with water. Comparing the cumulative beta characteristics of the two PM plots in table 2 shows that there is a small difference between Delaware-1 PM1, Delaware-2 PM1 and Delaware-2 PM2. Comparing the characteristics of the two RT plots shows a much larger difference between Delaware-1 and Delaware-2.

2.3 Willsboro

At the Willsboro site 4 wick pan samplers were installed; two in a no-tillage (NT) plot and 2 in a conventional tillage (CT) plot. Comparing the cumulative beta distribution curve characteristics of the plots with no-tillage and conventionally tillage shows there is difference between CT plot wick 4 and NT plots wick 1&2. The NT plot wick 1 had a larger portion of the water leached through preferential flow paths than compared with the NT plot wick 2 and CT plot wick 4. The beta distribution curve of CT plot wick 1 exhibits a higher heterogeneity than the other plots, possibly caused by preferential flow paths due to tillage.

2.4 Orchard

At the Orchard site, two plots were used: a mowed-grass-covered plot and a moss-covered-plot. Each plot had 1 wick and 1 gravity pan sampler installed. Comparing the cumulative beta distribution curve characteristics of the gravity and wick pan samplers for the grass covered plot as well as the moss covered plot is difficult, because the gravity pan sampler could overestimate the degree of soil heterogeneity as originally indicated by Boll^[1]. The HI and the IA show different results. For example, the grass covered plot using the gravity pan sampler results in a HI of 1.61 and an IA of 0.84. This HI is lower than the HI of the grass covered plot using the wick pan sampler (HI = 1.78), but the IA is higher than grass covered plot using the wick pan sampler (0.77).

Comparing the results of the HI and IA for the different sites is difficult, due to each site having different sampler techniques, different amounts and durations of applied water and lack of replication. Boll^[1] concluded that the results of solute samples (bromide

and blue dye) indicate that a large portion of the flow at the Freeville site and Delaware site occurred through matrix flow and that at the Orchard and Willsboro site more preferential flow occurred. In this paper the results of Orchard show higher values for HI and IA than for the other experiment sites, but the results of Delaware, Freeville and Willsboro do not show a significant difference. This paper for the first time presents a methodology to compare and contrast the level of heterogeneity in solute transport across several sites in the United States.

3 Conclusion

The recently introduced cumulative beta distribution technique is a versatile and simple technique of gaining valuable information regarding soil heterogeneity effects on solute transport in the vadose zone. The technique has been applied to 4 different sites across the USA. The resulting patterns from drainage and application of a conservative tracer to soils on these sites have been compared and contrasted. Significant differences in these patterns were observed even over a small spatial scale. These results therefore show that traditional sampling methods for solute transport, for example the gravity pan or suction lysimeters, or more recent inventions such as the multiple sample percolation systems may not be effective in estimating solute fluxes in soils when a significant degree of soil heterogeneity is present. Therefore, the cumulative beta distribution technique is also an excellent tool for guiding future decisions of experimental designs particularly in regard to the number of samples within one site and the number of sampling locations between sites required to obtain a representative field estimate. Ignoring soil heterogeneity in solute transport studies will likely result in under- or over- prediction of leached fluxes and potentially serious pollution of soils and/or groundwater may result.

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